

CD Burning Workstation — Cheat Sheet

Chain K — Hardware & Systems Foundations · one-page recall, not first-time learning

Blank Media & Drive

Task	Rule
Check burn capability	<code>wodim --devices</code>
CD-R vs CD-RW	CD-RW often unreadable in car stereos/older players — CD-R is the safer default
Capacity	650MB/74min or 700MB/80min

Acquisition & Conversion

Task	Command
Download audio only	<code>yt-dlp -x --audio-format <fmt></code>
Convert to Audio CD format	<code>ffmpeg -ar 44100 -sample_fmt s16 → WAV</code>
Exact Audio CD spec	44100Hz · 16-bit · 2-channel (stereo) PCM
Verify format before burning	<code>ffprobe</code>
Package managers	<code>apt</code> (Debian/Xubuntu) or <code>apk</code> (Alpine) — same tools either way

Burning & Verification

Task	Command / Rule
Burn Audio CD	<code>wodim -audio / cdrecord -audio</code>
Burn Data CD	<code>genisoimage / mkisofs → ISO</code>
No gaps between tracks	DAO (Disc At Once), not TAO
Trust a burn	Read-back/checksum verify — "completed" ≠ "good"
CD-DA data rate	176,400 bytes/sec ($44100 \times 16 \times 2 \div 8$)

Automation & Ethics

Rule	Detail
Script the pipeline	Validate args (URL list, disc size) + real exit codes, per Shell-And-Small-Software
Scope	Personal-use archival only — never redistribution
Avoid GUI weight	Brasero/K3B pull in GNOME/KDE libs — CLI is the recommended path on lightweight hardware

Chain K · CD-Burning-Workstation · full reasoning in LESSON.pdf